

A Transcriptomic Analysis of Rheumatoid Arthritis and Treatment Impact

Introduction

Rheumatoid arthritis (RA) is a chronic autoimmune disease that mostly affects joints, when immune system regulators attack joint tissue. The disease causes pain, swelling, stiffness, and loss of function in joints (1). It is important to start treatment of RA as soon as it is diagnosed to mitigate any potential joint damage which can lead to permanent disability. It is not fully known what causes RA; likely a combination of genes and exposure to environmental factors (1). The first line of treatment for RA is Methotrexate, which is a conventional disease-modifying anti-inflammatory drug (DMARD). It works to lower inflammation and slow an overactive immune system (2). Guidelines from the American College of Rheumatology strongly recommend methotrexate (MTX) alone (monotherapy) as the first treatment for people with moderate to high disease activity. Once response to MTX can be assessed, the patient may need to move on to add additional DMARDs or Biologics to reach remission or near-remission. 25-40% of patients respond to the first line of treatment, MTX monotherapy,(3) and published rates of remission vary greatly from 10%-60%(3).

The longer a patient diagnosed with RA is non responsive to treatment, the more joint damage will occur and could lead to the need for joint replacements or disability. Early treatment is essential. There is currently no standard omics approach to personalized treatment of RA, but this project is intended to explore differential gene expressions of healthy control samples, newly diagnosed (untreated) RA samples, and samples from various DMARD treated individuals to possibly find any gene targets or pathways initial treatments can target for individuals with certain gene expressions.

This project will take a metadata type approach of multiple studies with RA patients and omic effects of various treatments. The goal of this project is to compare differential gene expressions resulting from the most common treatment types of RA to determine efficacy and impact on expression to understand if treatment can begin with a more personalized approach based on genetic factors. This project may also identify currently untargeted disease mediators or genes. Study 1(5) contains RNA seq data from newly diagnosed RA patient's Peripheral blood monocytes (PBMCs) and their resulting RNA seq data after treatment with a Janus kinase inhibitor (JAK), upadacitinib. The published study released their RNAseq data for each experiment leg. The second study(6) contains RNAseq data from PBMCs of healthy individuals, newly diagnosed and untreated RA patients, and patients treated with MTX. The third study(8) contains RNAseq data from PBMCs of healthy individuals, individuals treated with MTX, and individuals treated with the biologic, Adalimumab (ADA), also known as the brand name, Humira.

Methods

Published studies were searched and selected on the NCBI Gene Expression Omnibus (GEO) portal to select data. RNAseq paired end data was imported to Galaxy via the SRA import tool from all studies as fastq files. TRIMMOMATIC was then run on the fastq files to trim low quality reads. The FastQC tool was used to check data quality- all reads were in the "good" range.

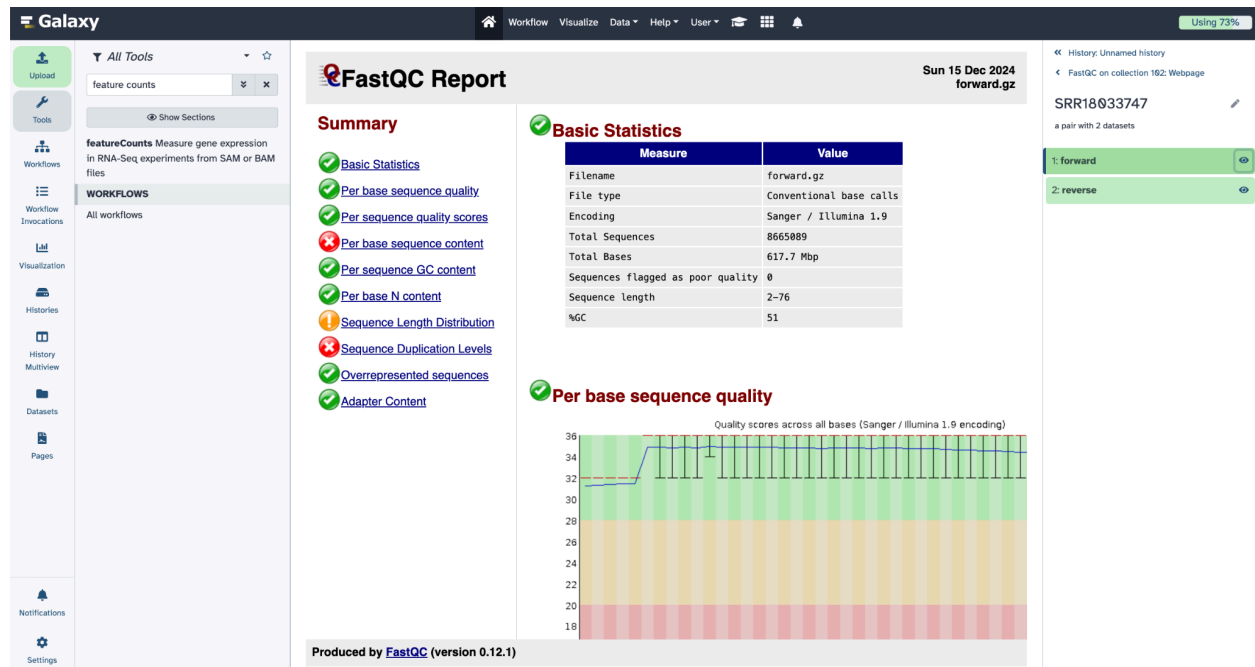


Figure 1: Galaxy view from sample SRR18033747, showing the FastQC Report and sequence quality

Then the reads were aligned to the human reference genome, Hg 38, using HiSTAT2 and converted to BAM files. From the alignments, FeatureCounts was finally used on the BAM files to create the gene count files used in R for differential gene expression analysis.

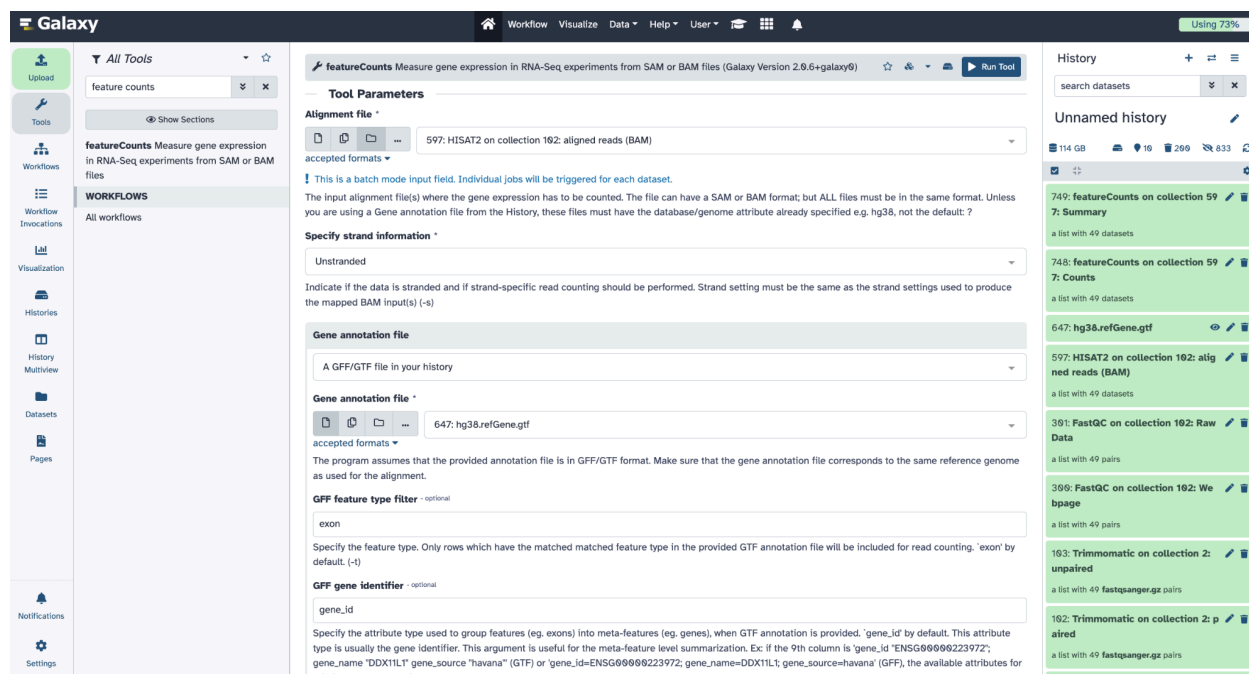


Figure 2: View of Galaxy. The middle console shows the settings used for FeatureCounts.

```
# We will import another set of data from RA positive individual's PBMC samples pre and post
treatment with MTX or a biologics ADA or etanercept (ETA) https://www.ncbi.nlm.nih.gov
/bioproject/PRJNA807144

# Load necessary libraries
library(dplyr)

# Get a list of CSV files in the 'data' directory
files <- list.files(path="/Users/Elizabeth/Documents/Capstone/Biologic and TNF blocker",
  pattern=".tabular", full.names=TRUE)

files
# Initialize an empty list to store data frames
data_list <- list()

# Loop over each file and read it into a data frame
for (file in files) {
  data <- read.delim(file)
  data_list[[file]] <- data
}

data_list
df<-data.frame(data_list)
df
```

Figure 3: Importing count files into R

Count data from all 3 studies were merged to create a dataframe used for the rest of the analysis.

```
mergeddata = cbind(JAKdata, MTXdata, ADA_data2)
#head(mergeddata)
head(mergeddata)
...
```

Description: df [6 × 23]

	PT1_postJAK <int>	PT2_postJAK <int>	PT1_postJAK <int>	healthy1 <int>	healthy2 <int>	healthy3 <int>	RA1pre <int>	RA2pre <int>	RA3pre <int>
DDX11L1	3	0	3	0	0	0	0	0	0
WASH7P	231	416	231	0	0	0	0	0	0
MIR6859-1	0	0	0	0	0	0	0	0	0
MIR6859-2	0	0	0	0	0	0	0	0	0
MIR6859-3	0	0	0	0	0	0	0	0	0
MIR6859-4	0	0	0	0	0	0	0	0	0

6 rows | 1-10 of 23 columns

Figure 4: Merged data frames of all count data to be used

R package DESEQ2 was used for the analysis. DESEQ2's differential expression tests are based on a negative binomial generalized linear model. We first needed to prep the data and create experimental group factors for the package. The first factor of “patient” includes patient identifiers. The second factor includes treatment type, where the control are healthy donors. The order in which the factors are listed are crucial, the patient must be in the order of it's corresponding treatment type. This is derived by the order in the count data frame.

```
# there could be many differences in library sized or sequencing depth. DEseq2 has it's own
# normalization method so we will be using that.

# We will now prep for DEseq2
# Create experiment group factors:

column_names = names(filterdata)
patient = c("PT1", "PT2", "PT3", "H1", "H2", "H3", "PT4", "PT5", "PT6", "PT7", "PT8", "PT9", "PT10",
"PT11", "PT12", "PT13", "PT14", "PT15", "PT16", "PT17", "PT18", "PT19", "PT20")
treatment = c("JAK", "JAK", "JAK", "aControl", "aControl", "aControl", "untreatedRA", "untreatedRA",
"untreatedRA", "MTX", "MTX", "MTX", "ADA", "ADA", "ADA", "ADA", "aControl", "aControl",
"aControl", "MTX", "MTX", "MTX", "MTX")
coldata = data.frame(patient = patient, treatment = treatment)
```

Figure 5: Preparing the experimental factors for DESEQ2.

A DESEQ2 object was created of the merged data to measure effect of treatment on the patients.

```

# Creating a DESeq2 object to measure effect of treatment on the patient

library(DESeq2)
dds <- DESeqDataSetFromMatrix(countData = filterdata,
                              colData = coldata,
                              design= ~treatment )
...

```{r}
dds
dds <- DESeq(dds)
...

```{r}
# Contrasts from ~treatment
resultsNames(dds)
...

[1] "Intercept"                "treatment_ADA_vs_aControl"    "treatment_JAK_vs_aControl"
[4] "treatment_MTX_vs_aControl"  "treatment_untreatedRA_vs_aControl"

...

```{r}
#Remove poorly expressed genes

keep<- rowSums(counts(dds)) >=5
dds<- dds[keep,]
...

```

**Figure 6:** DESeq2 object.

The results from DESeq2 were then plotted and further analyzed by a Gene Set Enrichment Analysis (GSEA) with ClusterProfiler and KEGG Gene Set Enrichment Analysis

```

```{r fig.height = 2, fig.width = 3}
# We will run a Gene Set Enrichment Analysis with ClusterProfiler
# we want the log2 fold change
original_gene_list <- res$log2FoldChange

# name the vector
names(original_gene_list) <- rownames(res)

# omit any NA values
gene_list<-na.omit(original_gene_list)

# sort the list in decreasing order (required for clusterProfiler)
gene_list = sort(gene_list, decreasing = TRUE)

gse <- gseGO(geneList=gene_list,
             ont ="ALL",
             keyType = "SYMBOL",
             nPerm = 10000,
             minGSSize = 3,
             maxGSSize = 800,
             pvalueCutoff = 0.05,
             verbose = TRUE,
             OrgDb = organism,
             pAdjustMethod = "none")
...

```

Figure 7: Setting up the Gene Set Enrichment Analysis

```
#Create gseKEGG object
kegg_organism = "hsa"
kk2 <- gseKEGG(geneList      = kegg_gene_list,
               organism      = kegg_organism,
               nPerm         = 10000,
               minGSSize     = 3,
               maxGSSize     = 800,
               pvalueCutoff  = 0.05,
               pAdjustMethod = "none",
               keyType       = "ncbi-geneid")
...
```

Figure 8: Setting up the gseKEGG object

Results and Discussion

The padj column in the DESEQ2 results table represents the p-value adjusted for multiple testing, and is the most important column of the results along with the log2foldchange. A threshold of padj < 0.05 and log2 fold change threshold of < -2 (for downregulated genes), >2 (for upregulated genes) was used to identify significant genes.

log2 fold change (MLE): treatment untreatedRA vs aControl

Wald test p-value: treatment untreatedRA vs aControl

DataFrame with 17398 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
DDX11L1	0.07882707	-0.445108	5.10051	-0.0872674	0.930459	0.938255
WASH7P	0.16579784	-0.445167	5.10051	-0.0872791	0.930450	0.938255
LOC729737	1.58467797	-0.445171	3.51511	-0.1266447	0.899222	0.938255
LOC102725121	0.00850365	-0.445130	5.10051	-0.0872718	0.930455	0.938255
TGIF1	3.09548532	-2.983361	2.04797	-1.4567431	0.145187	0.597516
...	...	...	...	...	...	...
LOC284898	0.605443	-1.130440	3.87420	-0.291787	0.77044959	0.9382552
PRKD3	8.475969	-5.728158	2.01021	-2.849537	0.00437829	0.0571155
MLC1	0.485469	0.516618	3.61464	0.142924	0.88635052	0.9382552
ZBTB41	0.585384	-0.445173	3.92276	-0.113485	0.90964641	0.9382552
TMEM181	3.005948	-3.836619	1.91294	-2.005615	0.04489733	0.2704804

out of 17397 with nonzero total read count

adjusted p-value < 0.1

LFC > 0 (up) : 2, 0.011%

LFC < 0 (down) : 1771, 10%

outliers [1] : 60, 0.34%

low counts [2] : 1, 0.0057%

(mean count < 0)

[1] see 'cooksCutoff' argument of ?results

[2] see 'independentFiltering' argument of ?results

Figure 9: DeSEQ2 results table

From the results table, we can see there are only 2 upregulated genes and 1771 downregulated genes when comparing untreated RA patients and the healthy control group. This is a great indicator that RA strongly downregulates any genes of interest, and we may further want to look for treatments to upregulate those genes.

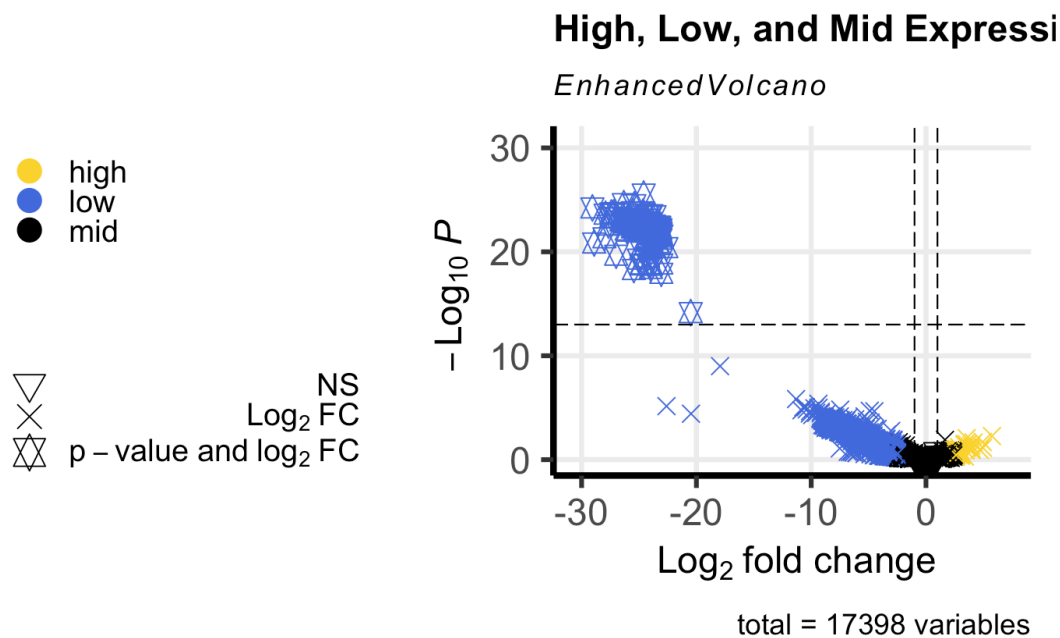


Figure 10: Enhanced Volcano plot of High, low and mid expressed genes.

For further visualization, the Volcano plot above confers with the results table, showing mainly down regulated genes. However, it does show that the most highly expressed genes are the ones that are upregulated.

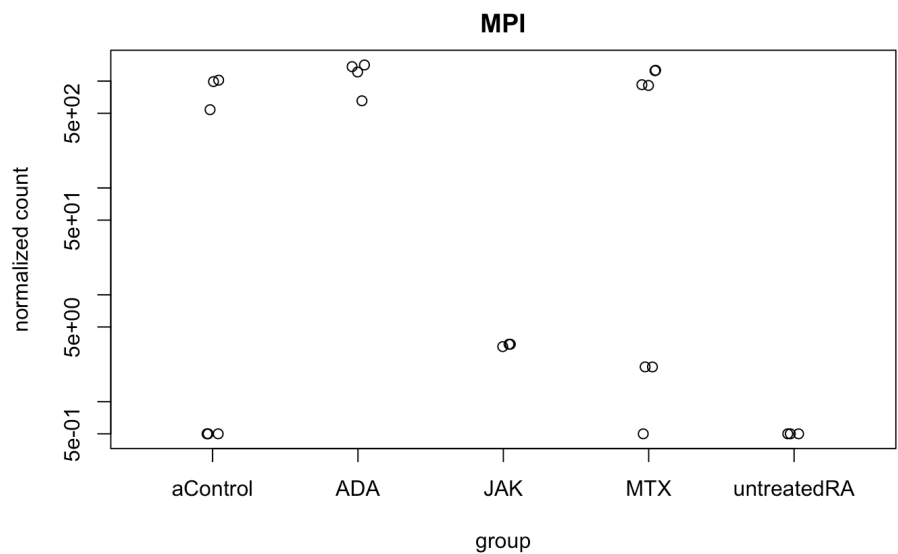


Figure 11: Gene, MPI, is the gene with the lowest adjusted p-value. The counts of each group are plotted.

We can see from the plot in Figure 11 that MPI has the lowest adjusted p-value, so the most differentially expressed gene out of all the groups. It is downregulated in untreated RA patients, but high in half of the control patients. All treatments do upregulate this gene, with ADA and MTX affecting it the most.

Before furthering investigation, result data quality was inspected using heatmaps and PCA.

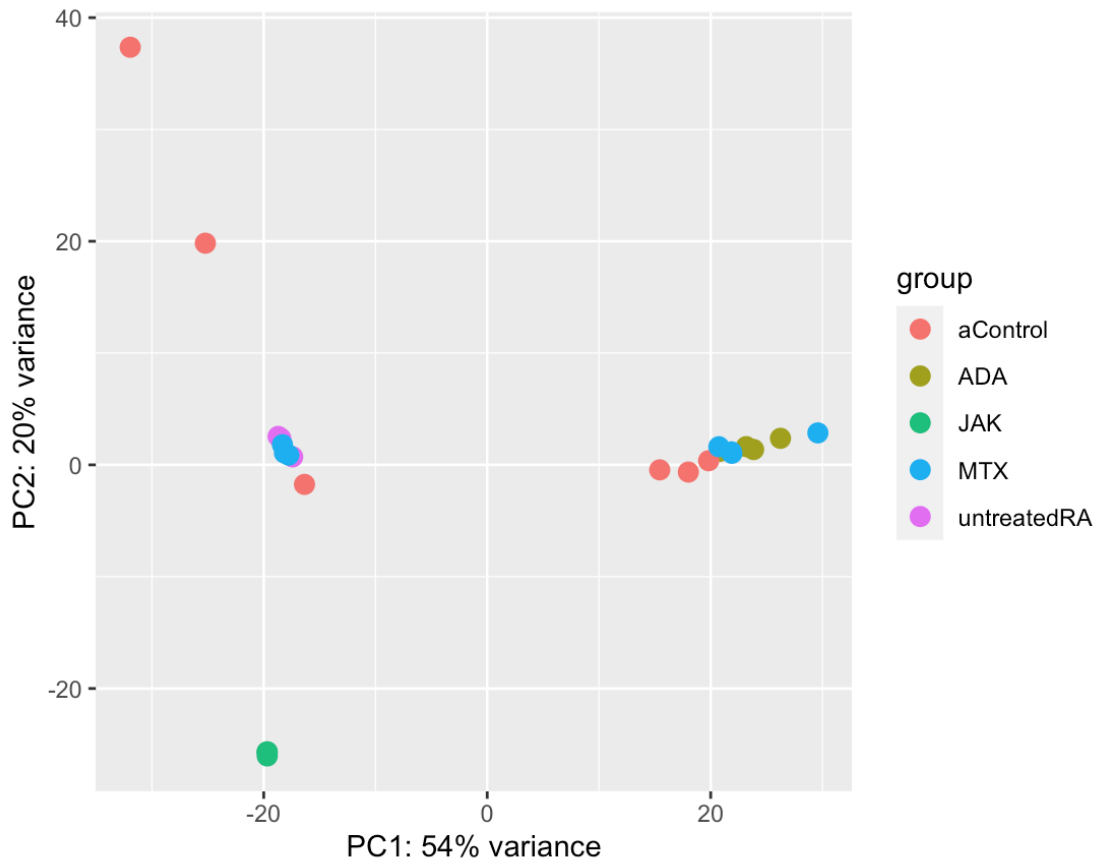


Figure 12: PCA of treatment groups

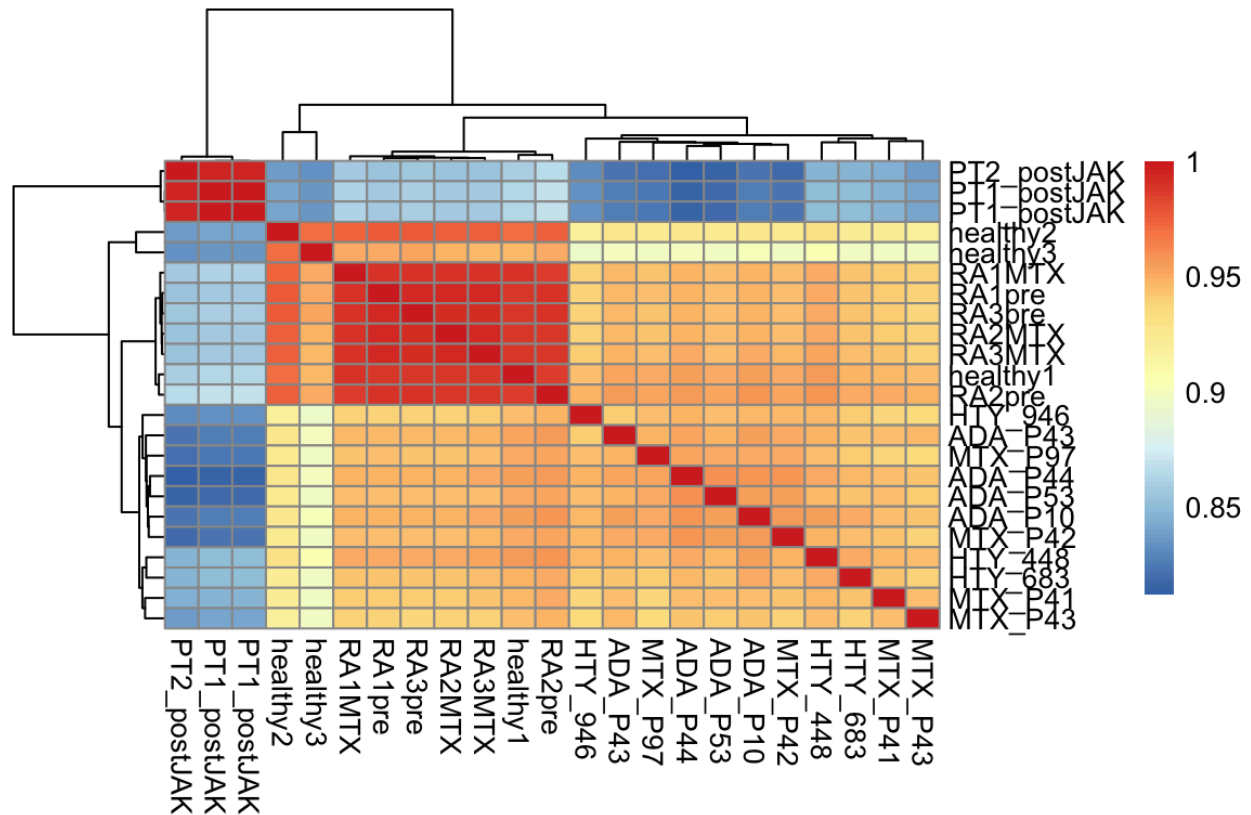


Figure 13: Heatmap of DeSEQ2 results to view quality of the samples

While we observe pretty high correlations across the board (> 0.999) in Figure 12, not all of the healthy groups have a high correlation, nor do the MTX groups. Unfortunately, there is not good separation amongst the treatment groupings in the PCA either, especially again in the control and MTX groups. Better quality data may be needed for more investigation.

Continuing on with this set of data, we can see what the genes that are most differentially expressed do within the body through Gene Set Enrichment Analysis and KEGG Pathway Analysis.

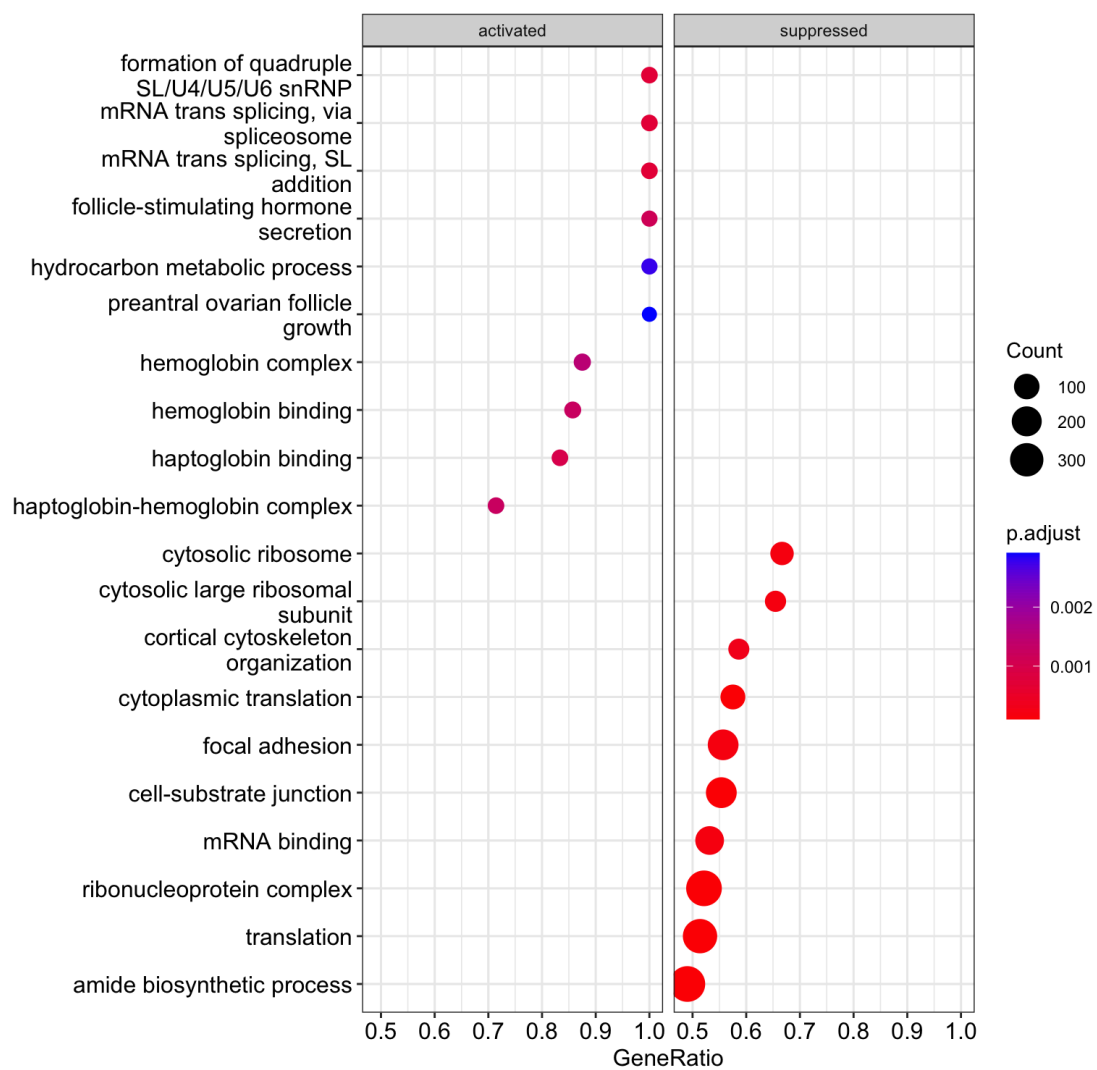


Figure 14: Dotplot of GSEA Results

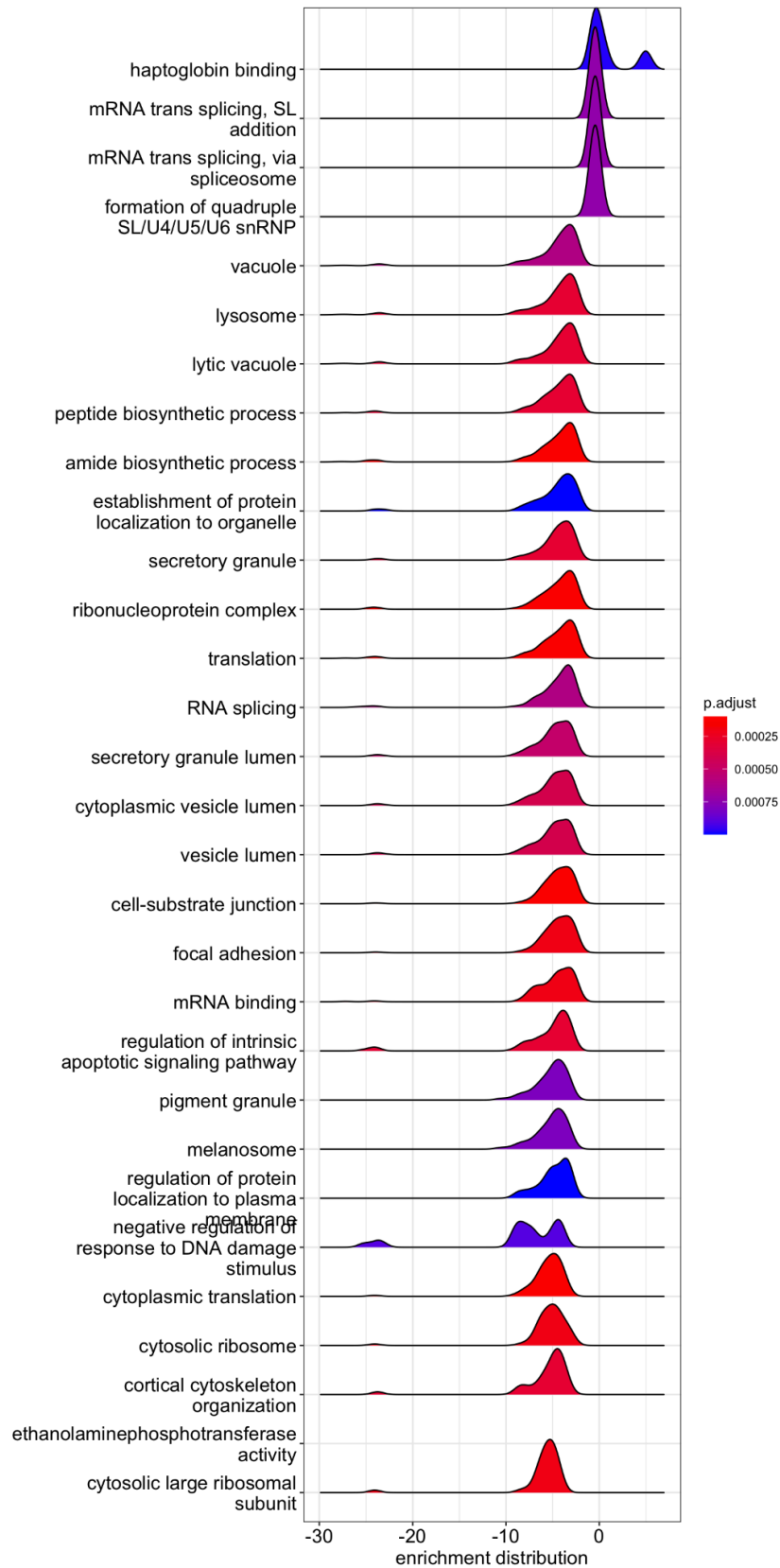


Figure 15: Ridgeplot of GSEA results

The most differentially expressed genes were suppressed in amide biosynthetic processes, translation, ribonucleoprotein complex, and mRNA binding.

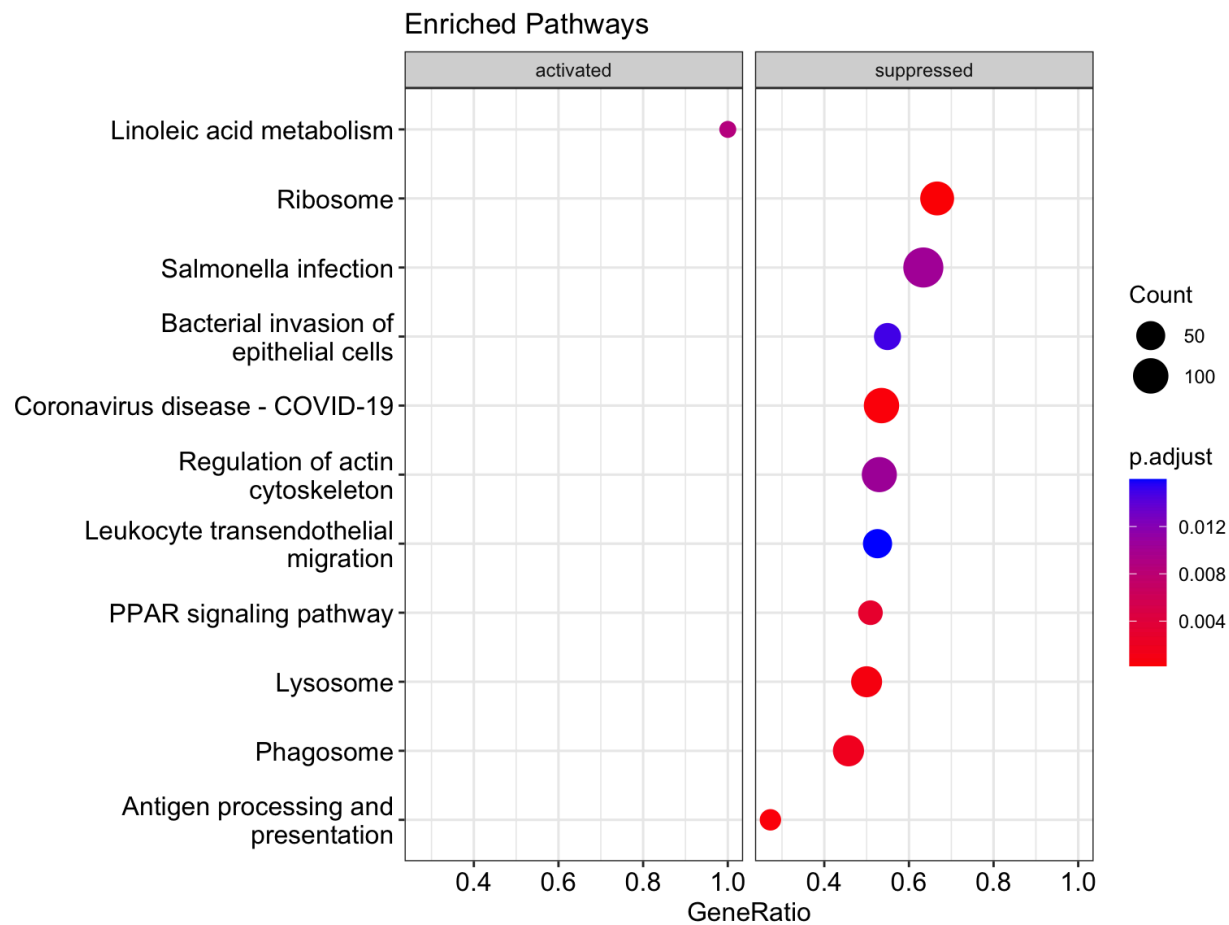


Figure 16: Dotplot of Kegg Enriched Pathways

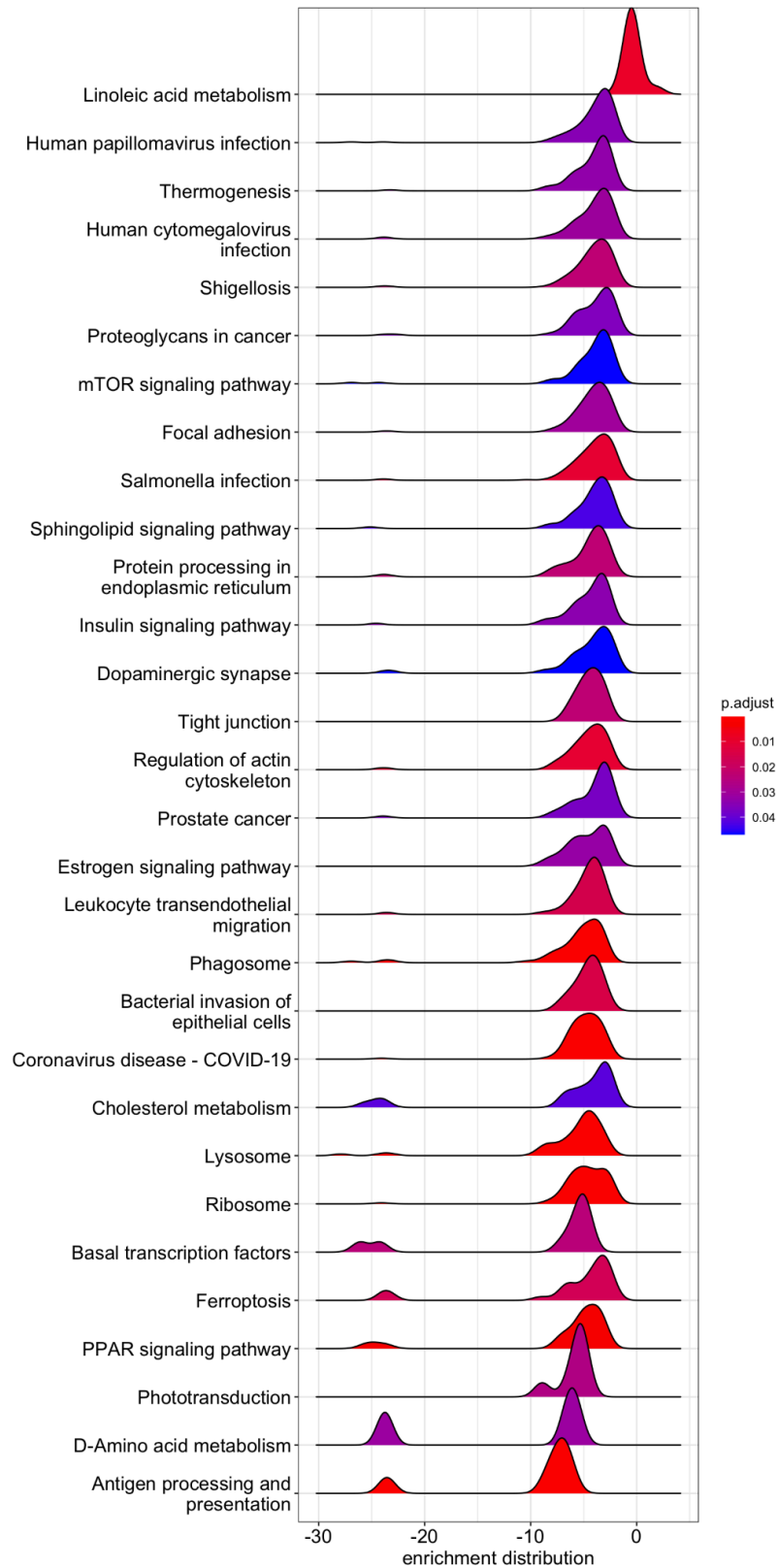


Figure 17: Ridgeplot of KEGG Enriched Pathways

Both analyses highlight that many processes are suppressed in RA and treatment, especially ribosomal and mRNA activity. The biosynthesis of ribosomes, the molecular machines that translate messenger RNA (mRNA) into proteins, is a fundamental biological process that is intimately linked to cell growth and proliferation.(11)

Conclusion

This project introduced me to handling multiple sets of data at a time, how to import them from published studies to bioinformatics tools such as Galaxy and how to look deeper into differentially expressed gene results using various gene enrichment analysis tools like KEGG, ClusterProfiler, and BioMart. It honed in on the concepts of differential gene expression and the statistical analyses used and why, such as adjusted p-values and log2 fold changes. Transcriptomic algorithms and tools used in the project unlock so much knowledge and data to explore all types of sequencing data and questions.

Based on the results of this project, RA strongly downregulates any genes of interest, and we may further want to look for treatments to upregulate those genes. Unfortunately the quality of the data was not optimal based on the results of DeSEQ2 shown on the heatmap and PCA. In future iterations I would choose more data and filter for quality. Based on the results, the ADA and MTX have more aligned results on gene expression than a JAK inhibitor. For future research, treatments that regulate ribosomal and mRNA activity should be further investigated.

References

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